

42.510 Statistical Learning for Data Science

Decision trees (Part II)

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Week 10, Lecture 2

Outline

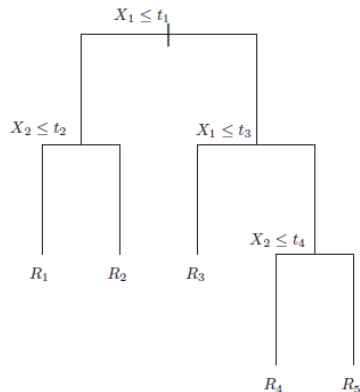
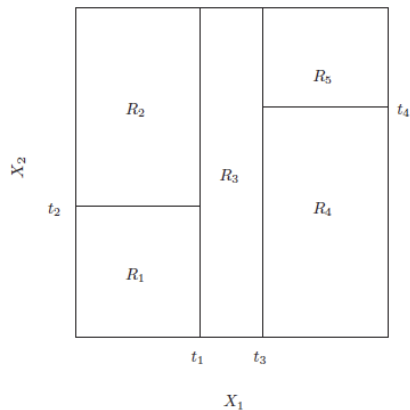
- Brief recap on CARTs
- Measures of impurity
- Pruning
- Trees versus linear models
- Advantages and Disadvantages of CARTs

Brief recap on CARTs

- Decision Trees can be applied to both regression and classification problems.
- The term **Classification And Regression Tree (CART)** is used to refer to procedures that learn either a *classification tree* or a *regression tree*.

Brief recap on CARTs

Structure of a Decision Tree



Brief recap on CARTs

Learning algorithm

Two main steps (regression):

- 1 Divide the predictor space into J distinct and non-overlapping regions $(R_1, R_2, \dots, R_J)$. This process uses **recursive binary splitting** and minimizes the RSS.

Brief recap on CARTs

Learning algorithm

Two main steps (regression):

- 1 Divide the predictor space into J distinct and non-overlapping regions $(R_1, R_2, \dots, R_J)$. This process uses **recursive binary splitting** and minimizes the RSS.
 - 2 For every observation that falls into the j -th region R_j , we make the same prediction c_j (mean of the response values for the training observations in R_j).
- `rpart` (in R) : stop if number of observations in a bucket goes below 20, or keep splitting if number of observations in a bucket $\geq 20/3 \approx 7$.

Perspective plot

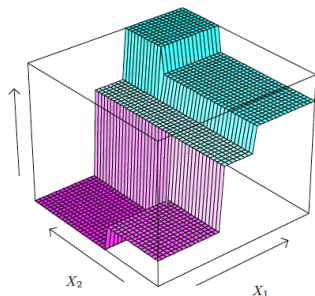


Figure 1: Perspective plot for a 2-dimensional prediction surface. (Source: James et al., 2014).

Brief recap on CARTs

Learning algorithm

For Classification Trees we use the same procedure, but:

- 1 We use a measure of impurity (instead of RSS) in the partitioning process.
- 2 The prediction c_j is the most commonly occurring class.

Classification Trees

Measures of impurity

- 1 Classification error rate (mis-classification rate):

$$E = 1 - \max_k(p_{mk})$$

where p_{mk} is the proportion of training observations in the m -th region that are from the k -th class.

Classification Trees

Measures of impurity (cont'd)

2 Gini index:

$$G = \sum_{k=1}^K p_{mk}(1 - p_{mk})$$

where K is the total number of classes, and G varies between 0 and 0.5.

Classification Trees

Measures of impurity (cont'd)

3 Cross-entropy:

$$D = - \sum_{k=1}^K (p_{mk} \log_2(p_{mk}))$$

Since $0 \leq p_{mk} \leq 1$, and D varies between 0 and 1.

Why use recursive binary splitting?

- Although we conduct exhaustive search at each step, total complexity is of order $J \times n \times p$ where
 - ▶ J = number of clusters
 - ▶ n = number of observations
 - ▶ p = number of variables
- Studies give guarantees for various specific structures.
- Is there any benefit in going beyond a binary split?
 - ▶ not much advantage
 - ▶ can also be achieved by multiple binary splits.
- Can we partition using linear combination of variables?
 - ▶ Yes, but interpretability goes down.

Back to R!

To learn CARTs, we will use the function `rpart`, implemented in the package `rpart`:

```
rpart(formula, data, method, control, ...)
```

Pruning

- The CART's learning algorithm is likely to build complex trees that overfit the training data.
- A smaller tree (with fewer region $R_1, R_2, \dots, R_J$) may lead to lower *variance* at the cost of a little *bias*.

Pruning

Idea:

- 1 Grow a large tree T_0 , and then,
- 2 *Prune* it back to obtain a *subtree*.

Some considerations:

- To determine how to prune the tree, we can use the *cross-validation error*.
- We cannot calculate the cross-validation error for all trees, because there are too many subtrees $\rightarrow$ it would take too long!

Pruning

Cost complexity pruning (or weakest link pruning)

Each value of a non-negative tuning parameter α corresponds to a sub-tree $T \subset T_0$ s.t. the value

$$\sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} (y_i - c_m)^2 + \alpha |T|$$

is as small as possible.

Note:

- When α increases, we pay a price for building a tree with many leaves.
- The above expression is a reminiscent of the LASSO, which controls the complexity of a linear model (first part of this course).

Pruning

(Full) algorithm for building a Regression Tree:

- 1 Use *recursive binary splitting* to grow a large tree on the training data.
- 2 Use *cost complexity pruning* to obtain a sequence of sub-trees as a function of α .
- 3 Use *k-fold cross-validation* to choose the best value of α .
- 4 Return the sub-tree from Step 2 that corresponds to the chosen value of α .

Pruning

How do we prune a Classification Tree?

We follow the same procedure, keeping in mind that the model error is calculated with a **measure of impurity**. This leads to the following expression

$$\sum_{m=1}^T E_m + \alpha |T|$$

which we still want to minimize.

Pruning with R!

The function `rpart` implemented in the package `rpart` already conducts pruning with a 10-fold cross-validation by default. `cp` denotes the cost-complexity parameter α . Check using:

```
printcp(model)
```

Then you may use:

```
prune(model, opt=cp)
```

Trees versus linear models

Let's compare a linear regression model

$$f(X) = \beta_0 + \sum_{j=1}^p \beta_j X_j$$

to a regression tree

$$f(X) = \sum_{m=1}^M c_m \cdot 1_{\{X \in R_m\}}.$$

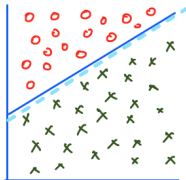
Trees versus linear models

Which model is better? The answer depends on the problem at hand.

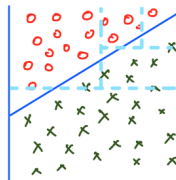
LOGISTIC REGRESSION

CART (CLASSIFICATION)

EXAMPLE 1

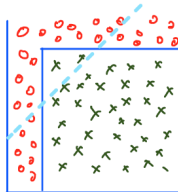


Linear model works well

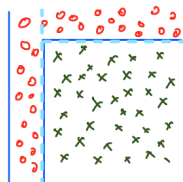


CART does not work well

EXAMPLE 2



Linear model does not work well



CART works well

Advantages and Disadvantages of CARTs

Pros:

- Interpretability
- Can be displayed graphically
- Can handle qualitative predictors (that take no continuous values)
- No assumptions on the relationship between input and output variables

Cons:

- They are not very accurate
- Not robust

References

- Martin et al. (2004) Competing approaches to predicting supreme court decision making. *Perspectives on Politics*, 2 (4), 761–767.
- James et al. (2023) *An Introduction to Statistical Learning: with Applications in R*, Chapter 8.1.